

VentureMiner AI

UI/UX Design Specification

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UI/UX Design Specification

Document 04 — VentureMiner AI The canonical design specification. Defines the design system, screen inventory, interaction patterns, and accessibility standards. Source of truth for every visual decision; deviations require a new revision.

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1. Purpose & Scope

This document is the design contract. It governs the look, feel, and behavior of every user-facing surface. Where this document disagrees with code, this document wins until the next revision.

1.1 In scope

Design tokens (color, typography, spacing, motion)

- Accessibility and responsive strategy.
- Visual states (empty, loading, error).

1.2 Out of scope

- Marketing site (handled by Web team outside this suite).
- Mobile native apps (v2+).
- Print design (handled inside reports via Document 09).

2. Design Principles

1. Clarity over cleverness. Every screen must answer "what is this and what do I do?" in < 5 seconds.
2. Evidence, not assertion. Visual hierarchy favors source citations and traces.
3. Operator-grade density. A senior researcher must be able to scan a portfolio in one screen.
4. Calm by default. The system is research-focused; visual noise is a defect.
5. Bias to legibility. Body text $\geq 14\text{px}$; never sacrifice readability for compactness.

3. Brand & Voice

3.1 Voice

- Authoritative but humble. State conclusions with confidence; cite the evidence.
- Specific over vague. Prefer "12,400 monthly searches" to "lots of interest".
- Operator's language. Use domain terms (TAM, GTM, CAC) without dumbing them down.

3.2 Tone spectrum

Empty states	Encouraging, instructive
Error states	Apologetic, actionable
Success	Brief, factual
Reports	Analytical, neutral

4. Information Architecture

- Home (/)
- Marketing (one-pager)
- App (/app)
- Onboarding (/app/onboarding)
- Dashboard (/app/dashboard)
- Discovery (/app/discovery)
- Search (/app/discovery/search)
- Trends (/app/discovery/trends)
- Pains (/app/discovery/pains)

- Opportunities (/app/opportunities)
- ■■■ List (/app/opportunities)
- ■■■ Detail (/app/opportunities/:id)
- ■■■ Compare (/app/opportunities/compare)
- Validations (/app/validations)
- ■■■ Detail (/app/validations/:id)
- Reports (/app/reports)
- ■■■ Briefs (/app/reports/briefs)
- ■■■ Full (/app/reports/full)
- ■■■ Compare (/app/reports/compare)
- Portfolio (/app/portfolio)
- Watchlists (/app/watchlists)
- Settings (/app/settings)
- Workspace
- Members
- Rubric
- Sources
- Integrations
- API tokens
- Billing
- Audit log

5. Navigation Model

- Top nav (persistent): logo · workspace switcher · global search (Cmd-K) · notifications · avatar.
- Left rail (contextual): app sections.
- Breadcrumb (deep pages): workspace › section › page.
- Quick action bar (bottom): floating "Run Discovery" / "Validate" / "Generate Brief".

5.1 Global search (Cmd-K)

- Indexes: opportunities, reports, sources, watchlists, members, settings pages.
- Keyboard-first; results in < 50ms p75.
- Recent + suggested surfaced above results.

6. Design Tokens

Tokens are stored in a JSON file (exported to CSS variables and JS).

6.1 Token structure

```
{
  "color": { "primary": { "50": "...", "500": "...", "900": "..." } },
  "font": { "size": { "xs": 12, "sm": 14, "base": 16, "lg": 18, "xl": 20, "2xl": 24,
    "3xl": 30, "4xl": 36 } },
  "space": { "0": 0, "1": 4, "2": 8, "3": 12, "4": 16, "5": 20, "6": 24, "8": 32,
    "10": 40, "12": 48, "16": 64 },
```

```
"radius":{ "sm": 4, "md": 8, "lg": 12, "xl": 16, "full": 9999 },
"shadow":{ "sm": "...", "md": "...", "lg": "..."}
}
```

7. Typography

Display	Inter	36 / 30	700	1.1
H1	Inter	24	700	1.25
H2	Inter	20	600	1.3
H3	Inter	18	600	1.4
Body	Inter	16	400	1.5
Small	Inter	14	400	1.5
Mono	JetBrains Mono	14	400	1.5

- Minimum body size: 14px (small), 16px (body). Reports may use 11px for footnotes.
- Long-form report body: 16–17px, 1.6 line-height for reading.

8. Color System

8.1 Brand palette

primary-500	#2563EB	Primary actions, links
primary-600	#1D4ED8	Hover
primary-700	#1E40AF	Pressed

8.2 Semantic palette

success	#10B981	#34D399	Pass, positive
warning	#F59E0B	#FBBF24	Caution
danger	#EF4444	#F87171	Error, destructive
info	#3B82F6	#60A5FA	Informational

8.3 Neutral palette (slate)

slate-50	#F8FAFC
slate-100	#F1F5F9
slate-200	#E2E8F0
slate-300	#CBD5E1
slate-400	#94A3B8
slate-500	#64748B

slate-50	#F8FAFC
slate-600	#475569
slate-700	#334155
slate-800	#1E293B
slate-900	#0F172A

8.4 Score color scale (used in scoring UI)

0.0 – 3.0	#EF4444 (red)
3.0 – 5.5	#F59E0B (amber)
5.5 – 7.5	#10B981 (green)
7.5 – 10	#0EA5E9 (sky)

9. Iconography

- Set: Phosphor (regular weight default; bold for emphasis).
- Size: 16px (inline), 20px (UI), 24px (feature), 32px (illustrations).
- Color: inherits from text by default; never rainbow-coded.

10. Spacing & Layout Grid

- Base unit: 4px.
- Common spacing: 4, 8, 12, 16, 20, 24, 32, 40, 48, 64.
- Container max-width: 1440px (app), 720px (read view).
- Grid: 12 columns, 24px gutter, 16px outer margin (≥ 768px viewport).
- Breakpoints: sm 640 / md 768 / lg 1024 / xl 1280 / 2xl 1536.

11. Components Library

The component library is implemented in React + Radix + Tailwind. Each component has:

- A contract (props, behavior, accessibility).
- States (default, hover, focus, active, disabled, loading, error).
- Storybook story.
- Tests (unit + visual).

11.1 Core components

Button	primary, secondary, ghost, danger; size sm/md/lg	Disabled / loading / icon-only
Input	text, number, password, search, url	Error state with inline help

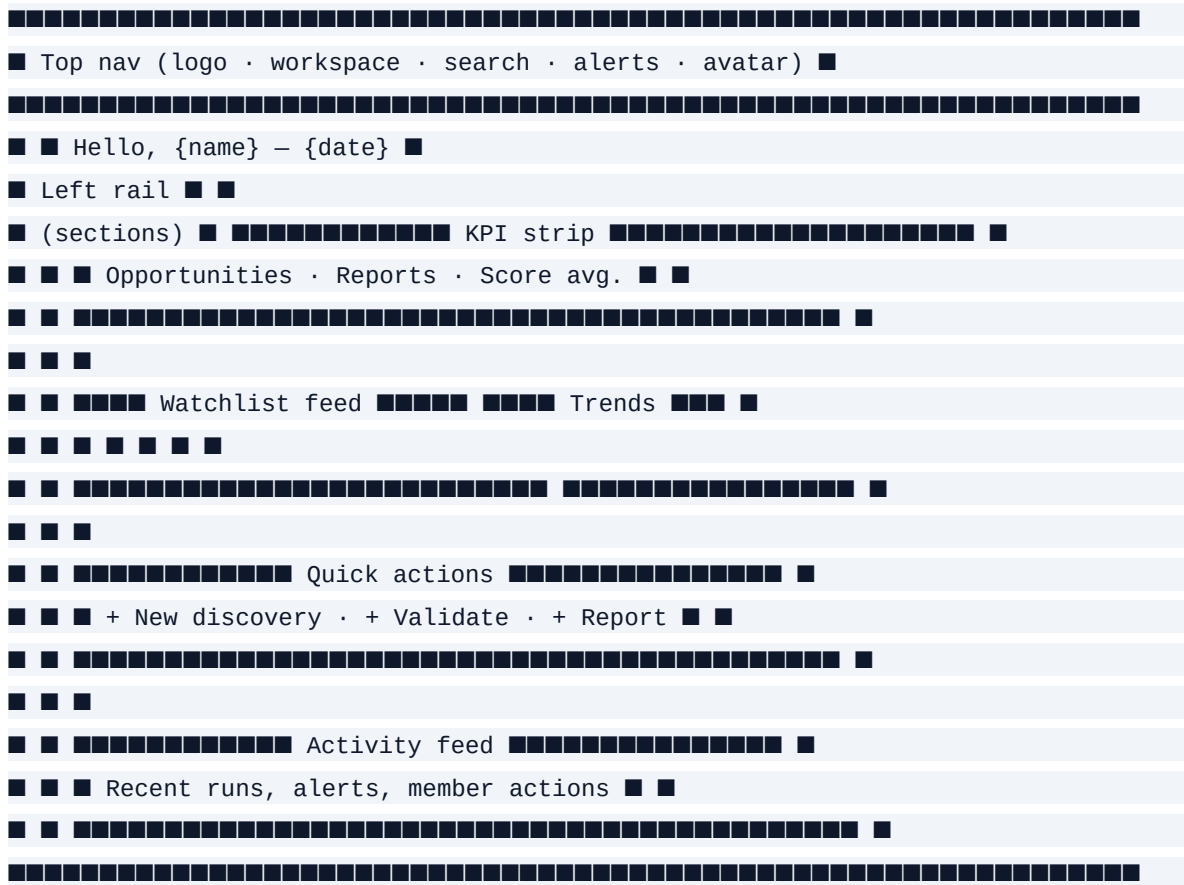
Button	primary, secondary, ghost, danger; size sm/md/lg	Disabled / loading / icon-only
Textarea	auto-resize	
Select	single, multi, async, combobox	Headless via Radix
Combobox	search + multi	Used for tags, sources
DateRangePicker	presets + custom	
Tabs	line, pill	
Table	sortable, filterable, paginated, virtualized	Sticky header
Card	default, outlined, elevated	
Modal	sm/md/lg/xl; destructive variant	Focus-trapped
Drawer	right, bottom (mobile)	
Toast	success, info, warning, error	Auto-dismiss 4s
Tooltip	light/dark; top/bottom/left/right	
Popover		
Dialog		Modal variant
Accordion	single, multi	
Avatar	with status dot	
Badge	tone variants	
Breadcrumb		
Pagination		
EmptyState	illustration + CTA	
LoadingSkeleton	line, block, table	
ProgressBar	determinate, indeterminate	
ScoreBadge		Color by range
CitationPopover		Shows source on hover
EvidencePanel		Lists cited evidence
RubricEditor		Drag-and-drop dimensions
ReportPreview		Read-only render
Alert	banner	

SCR-AUTH-001	Sign in	/app/signin	Public
SCR-AUTH-002	Sign up	/app/signup	Public
SCR-AUTH-003	Email verification	/app/verify	Public
SCR-AUTH-004	MFA challenge	/app/mfa	Partial
SCR-OB-001	Onboarding tour	/app/onboarding	Auth
SCR-OB-002	Sample opportunity	/app/onboarding/sample	Auth
SCR-DASH-001	Dashboard	/app/dashboard	Auth
SCR-DISC-001	Discovery search	/app/discovery/search	Auth
SCR-DISC-002	Discovery results	/app/discovery/search/:runId	Auth
SCR-DISC-003	Trends dashboard	/app/discovery/trends	Auth
SCR-DISC-004	Pain feed	/app/discovery/pains	Auth
SCR-OPP-001	Opportunity list	/app/opportunities	Auth
SCR-OPP-002	Opportunity detail	/app/opportunities/:id	Auth
SCR-OPP-003	Compare	/app/opportunities/compare	Auth
SCR-VAL-001	Validation detail	/app/validations/:id	Auth
SCR-SCORE-001	Rubric list	/app/settings/rubric	Admin
SCR-SCORE-002	Rubric editor	/app/settings/rubric/:id	Admin
SCR-RPT-001	Brief viewer	/app/reports/briefs/:id	Auth
SCR-RPT-002	Full report viewer	/app/reports/full/:id	Auth
SCR-RPT-003	Compare report	/app/reports/compare/:id	Auth
SCR-PORT-001	Portfolio	/app/portfolio	Auth
SCR-WATCH-001	Watchlists	/app/watchlists	Auth
SCR-WS-001	Workspace settings	/app/settings/workspace	Admin
SCR-WS-002	Members	/app/settings/members	Admin
SCR-INT-001	Integrations	/app/settings/integrations	Admin
SCR-API-001	API tokens	/app/settings/api	Admin
SCR-BILL-001	Billing	/app/settings/billing	Admin
SCR-AUDIT-001	Audit log	/app/settings/audit	Admin (Ent)
SCR-ERR-001	404	/app/404	Public
SCR-ERR-002	500	/app/500	Public

13.1 SCR-DASH-001 — Dashboard

Purpose: Give a returning user a 5-second read on what changed and what to do next.

Layout:



Behavior:

- KPI strip links to filtered lists.
- Trends panel shows top 5 trends in user's watched categories.
- Activity feed paginates 20 per page; infinite scroll on mobile.

States:

- Empty (new user): 3 sample opportunities CTA.
- Loading: skeleton blocks.
- Error: retry banner + reload.

Accessibility:

- All sections have headings (h2).
- All KPIs are links with descriptive aria-label.
- Color is not the only signal for score color-coding — value always text.

13.2 SCR-OPP-002 — Opportunity detail



Title · Status · Score badge · Actions (Validate, Brief)

Tabs

Overview · Validation · Report · Notes · History

Overview tab

Key facts

Score breakdown

Market size, growth,

Dimensions, weights,

demand, GTM, risks

sub-scores, rationale

Evidence panel

Cited snippets, sources, freshness

Behavior:

- Tabs are deep-linkable (?tab=validation).
- Evidence panel shows citations as hover popovers and a clickable list.
- Score breakdown is expandable per dimension.

Accessibility:

- Tabs use ARIA tabs pattern.
- Each evidence item has a screen-reader-readable source.

13.3 SCR-OPP-003 — Compare

- A 2-up layout by default; up to 4 opportunities per screen.
- Sticky header with scores.
- Highlight differing cells in amber.
- "Generate comparison report" CTA.

14. Interaction Patterns

14.1 Destructive actions

- Always confirm via modal with explicit "Type the title to confirm" for hard deletes.
- Provide a 5-second "undo" toast where state allows.

14.2 Async runs (Discovery, Validation, Report)

- Inline progress with step list.
- SSE-driven; falls back to polling every 5s on error.
- User can navigate away; result delivered to in-app + email.

• User can cancel

14.3 Long-form content

- Reports use a separate "read view" with serif option (toggle).
- Footnotes are inline (hover) and a sidebar list.

14.4 Bulk actions

- Selection state in lists; bulk bar appears at top.
- Confirm for bulk destructive.

14.5 Cmd-K

- Always available; index updated on view.

15. Empty / Loading / Error States

15.1 Empty state template

[Illustration]

Headline (1 line)

Subhead (1-2 lines, what to do)

Primary CTA

Secondary link

15.2 Loading

- Skeletons for predictable layouts.
- Spinner for actions.
- Progress for long-running tasks (with ETA when known).

15.3 Error

- Inline for field errors; banner for page errors; full-page for 500s.
- Always includes a "what to do next" — retry, contact support, return.

16. Motion & Microinteractions

- Durations: fast 120ms, default 200ms, slow 320ms.
- Easing: cubic-bezier(0.2, 0.8, 0.2, 1) (standard), cubic-bezier(0.4, 0, 0.2, 1) (decelerate).
- Use motion for: state transitions, focus changes, score updates.
- Avoid motion for: content shuffling during reading, distracting micro-loops.
- Reduced motion: respect prefers-reduced-motion.

17. Accessibility

- WCAG 2.1 AA is the minimum.

- Visible focus ring (2px solid primary-500).
- Color contrast: $\geq 4.5:1$ body, $\geq 3:1$ large text, $\geq 3:1$ UI components.
- Form fields have labels (no placeholder-only).
- Errors announced via `aria-live="polite"`.
- Tables use and captions.
- Modals trap focus and restore on close.
- Charts have a text alternative (table or summary).

18. Responsive Strategy

- Mobile-first.
- Breakpoints: sm 640 / md 768 / lg 1024 / xl 1280.
- App is desktop-first (≥ 1024); below that, it is a read + quick-action view.
- Reports are responsive (single column on mobile, multi-column on desktop).
- Tables collapse to stacked cards on < 768 .

19. Internationalization

- All strings externalized (t('...')).
- Date/number/currency via Intl.
- No hard-coded text in components.
- Right-to-left deferred to v3.

20. Dark Mode

- Default to system preference; user override in settings.
- Tokens designed to swap automatically.
- Charts re-palette for dark mode.
- Avoid pure black; use slate-900 background.

21. Performance Budgets (visual)

See TRD §8.3 for the page-level JS/CSS budgets. The visual budget is:

- No layout shift > 0.1 CLS.
- LCP $< 2.5s$ p75.
- TTI $< 3.0s$ p75 on dashboard.

22. Design Quality Checklist

- ☐ States (default, hover, focus, active, disabled, loading, error) implemented.
- ☐ Keyboard navigable.
- ☐ Screen reader tested.
- ☐ Color contrast verified.
- ☐ Reduced motion respected.
- ☐ Empty / loading / error states designed.
- ☐ Storybook story added.
- ☐ Visual regression test added.
- ☐ Responsive at sm/md/lg/xl verified.
- ☐ Dark mode verified (if supported).

23. Appendix

23.1 Component-to-screen matrix (excerpt)

ScoreBadge	Dashboard, Opportunity list, Compare, Report
CitationPopover	Opportunity detail, Report, Validation
RubricEditor	Settings
ReportPreview	Report viewer, Compare report

23.2 Revision history

v0.5	2026-07-20	Doc Team	All sections drafted
v1.0	2026-07-20	Doc Team	First approved version

23.3 Cross-references

- Application flows: Document 03.
- Accessibility policy: Document 21 (Security).
- Performance: Document 02 §8.

End of Document 04 — UI/UX Design Specification. Tokens are the source of truth; any new value must be added to the token system first.